THE UNIVERSITY OF BRITISH COLUMBIA
Faculty of Forestry

Dear Dr. Wake,

Please consider our manuscript, entitled "Overlooked model uncertainties may misinform forest management strategies", for publication in *Nature Climate Change*. This manuscript leverages methods from climatology to decompose ecological forecasts and guide forest management under uncertainty.

Forests are showing signs of becoming sources of carbon, putting pressure on ecological forecasts to inform current management decisions. Misinformed decision-making—because of incomplete forecasts—could have long-term detrimental consequences for ecosystems[1,2], driving forest decline and reducing carbon storage. Yet, current studies often rely on a narrow set of models and ignore a large part of the uncertainties[3,4], limiting their practical value for forest management.

Here, we apply climatological methods designed to aid decision-making given many models and associated uncertainties. We considered over 1,350 projections of tree species range shifts, using diverse ecological and climatological models, and across different emissions scenarios. By fully encompassing the different sources of uncertainty, we were able to quantify the contribution of each source to the forecast uncertainty, across species and across biomes.

Our results show that ecological models represent the largest source of uncertainty—up to two thirds—even under vastly different emission scenarios. While previous studies have strongly underestimated the overall forecast uncertainty, our workflow provides a comprehensive view of potential futures for forests. We also identify regions where management could take immediate action with a low risk of failure, as well as regions where models strongly disagree, for which uncertainty management and diversification of options would be necessary.

Using climatological methods to partition sources of forecast uncertainty, our findings could reshape the debate on ecological forecasting of future species ranges. Although ecologists have spent decades debating the types of ecological model, our results provide a path to guide forest policy and management as science is advancing. We advocate the systematic incorporation of uncertainties into ecological forecasts and decision-making processes[1,2,5], and suggest that managers should internalize ecological uncertainty through diversified and more risk-averse strategies.

All authors contributed to this work and approve this version for submission. The manuscript is 2540 words, with 61 references and 4 figures, and is not under consideration elsewhere. We hope you find it suitable for publication in *Nature Climate Change*, and look forward to hearing from you.

Thank you in advance for your consideration.

Sincerely,

Victor Van der Meersch, PhD
Forest & Conservation Sciences
University of British Columbia

References

[1] T. P. Dawson et al. *Science* (2011).

[2] M. C. Urban et al. *Science* (2016).

[3] J. Wessely et al. *Nature Ecology & Evolution* (2024).

[4] M. Hanewinkel et al. *Nature Climate Change* (2013).

[5] A. Saltelli et al. *Nature* (2020).